Answer key

SAT Practice Test 3, Reading and Writing

Module 1

NO.	ANSWER	SKILL
1	C	Words in context
2	D	Words in context
3	B	Words in context
4	A	Words in context
5	A	Words in context
6	B	Text structure and purpose
7	D	Text structure and purpose
8	C	Text structure and purpose
9	A	Cross-text connections
10	C	Central ideas and details
11	B	Central ideas and details
12	D	Command of evidence: quantitative
13	C	Command of evidence: quantitative
14	A	Command of evidence: textual
15	B	Command of evidence: textual
16	D	Command of evidence: textual
17	C	Inferences
18	B	Inferences
19	D	Inferences
20	A	Form, structure, and sense
21	A	Boundaries
22	C	Form, structure, and sense
23	B	Form, structure, and sense
24	D	Boundaries
25	D	Boundaries
26	A	Form, structure, and sense
27	B	Form, structure, and sense
28	C	Transitions
29	D	Transitions
30	C	Transitions
31	B	Rhetorical synthesis
32	A	Rhetorical synthesis
33	D	Rhetorical synthesis

Module 2

NO.	ANSWER	SKILL
1	D	Words in context
2	B	Words in context
3	C	Words in context
4	A	Words in context
5	C	Words in context
6	D	Text structure and purpose
7	A	Text structure and purpose
8	B	Text structure and purpose
9	D	Cross-text connections
10	B	Central ideas and details
11	A	Central ideas and details
12	C	Command of evidence: quantitative
13	C	Command of evidence: quantitative
14	D	Command of evidence: textual
15	A	Command of evidence: textual
16	B	Command of evidence: textual
17	B	Inferences
18	C	Inferences
19	A	Inferences
20	D	Form, structure, and sense
21	A	Boundaries
22	D	Form, structure, and sense
23	B	Form, structure, and sense
24	C	Form, structure, and sense
25	D	Form, structure, and sense
26	A	Boundaries
27	C	Boundaries
28	B	Transitions
29	C	Transitions
30	A	Transitions
31	D	Rhetorical synthesis
32	B	Rhetorical synthesis
33	A	Rhetorical synthesis

Answer key

SAT Practice Test 3, Math

Module 1

NO.	ANSWER	SKILL
1	**C**	Percentages
2	**B**	Data analysis
3	**A**	Linear equations
4	**D**	Linear equations in context
5	**D**	Systems of equations
6	**B**	Functions
7	**C**	Nonlinear expressions
8	**A**	Ratios and rates
9	**C**	Triangles
10	**D**	Circles
11	**B**	Quadratics
12	**A**	Probability
13	**C**	Statistics
14	**B**	Scatterplots
15	**D**	Inequalities
16	**A**	Functions
17	**B**	Exponential functions
18	**C**	Linear functions
19	**D**	Percent change
20	**A**	Right triangles
21	**20**	Percent change
22	**6**	Ratios and rates
23	**396**	Volume
24	**12**	Linear equations
25	**23**	Systems of equations
26	**63**	Area
27	**11**	Quadratics

Module 2

NO.	ANSWER	SKILL
1	**B**	Volume
2	**A**	Probability
3	**D**	Scatterplots
4	**C**	Functions
5	**C**	Statistics
6	**D**	Linear equations
7	**B**	Percentages
8	**A**	Circles
9	**D**	Exponential functions
10	**B**	Circles in the plane
11	**A**	Exponential equations
12	**C**	Quadratics
13	**A**	Systems of equations
14	**C**	Absolute value
15	**B**	Equivalent expressions
16	**D**	Ratios and rates
17	**B**	Statistics
18	**C**	Equivalent expressions
19	**A**	Functions
20	**D**	Percentages
21	**5**	Linear equations
22	**-1**	Quadratics
23	**10**	Quadratics
24	**5**	Circles in the plane
25	**34**	Similar triangles
26	**8**	Quadratics
27	**-1**	Systems with nonlinear equations

Score yourself. Count the questions you got right in each module.

Reading and Writing, Module 1: ___ of 33

Module 2: ___ of 33

Math, Module 1: ___ of 27

Module 2: ___ of 27

Answer explanations

SAT Practice Test 3, Reading and Writing, Module 1

Questions 1 to 17

1 **Choice C** Words in context

The colon explains what she does: she lets each loaf rise a full day however busy the shop is. That shows patience. Haste is the opposite, and nothing indicates secrecy or luck.

2 **Choice D** Words in context

Each side gave up something, which defines a compromise. A rivalry or stalemate would mean no agreement, and a monopoly is unrelated.

3 **Choice B** Words in context

Her earlier style contrasts with the short, plain lines of the new book, so the earlier sentences were elaborate and ornate. Terse and plain describe the new style, and hesitant does not fit elaborate.

4 **Choice A** Words in context

Birds that show little fear of approaching humans are unwary. Aggressive, migratory and ravenous do not explain the lack of fear.

5 **Choice A** Words in context

The board began to take her ideas seriously only after a smaller version succeeded, so the proposal had first been dismissed as impractical. Profitable and dazzling would not lead to dismissal, and a routine proposal would not be dismissed.

6 **Choice B** Text structure and purpose

The underlined sentence says what each partner provides, which explains how the partnership works. It does not dispute an earlier claim, compare organisms, or describe researchers.

7 **Choice D** Text structure and purpose

Counting tiles and crushing the sheet music suggest Ravi's tension and disappointment. The text gives no reasons from the director, shows no relief, and does not discuss why he became a musician.

8 **Choice C** Text structure and purpose

The first stanza describes an orchard at the season's end, and the second attributes belief to the ladder. No list of tasks, childhood or argument appears.

9 **Choice A** Cross-text connections

Text 2 suggests extra support at home may explain the scores, so the studies do not prove homework is the cause. The teacher does not say homework hurts or recommend more of it.

10 **Choice C** Central ideas and details

The text describes replay of daily activity during sleep and better recall after sleeping, so sleep helps consolidate learning. It contradicts the idea of an inactive brain, and nothing suggests sleep matters less than studying.

11 **Choice B** Central ideas and details

The text contrasts the village's look of stillness with a record of fires, a shifting river and emigration, and calls the stillness the result of recoveries. It does not predict disappearance, or say anything about visitors' reading or the mill's success.

12 **Choice D** Command of evidence: quantitative

Rapid growth in spring is shown by the rise from 120 in March to 310 in May, which the first choice reports. The others reverse the trend, list extremes without the rise, or cover a slower period.

13 **Choice C** Command of evidence: quantitative

The claim links iced drinks with warm weather, so the right choice compares the iced share on warm and cool days. The others discuss hot drinks or lack the comparison.

14 **Choice A** Command of evidence: textual

The change is shown by the later date of mid-March compared with the first week of April, and only the first quotation gives the later arrival. The others describe the record keeping or the earlier date alone.

15 **Choice B** Command of evidence: textual

Higher attention scores in the group that practised an instrument support the claim. The age of the children and the length of the lessons do not show an effect on attention.

16 **Choice D** Command of evidence: textual

The text says the deck swayed in strong winds and unsettled walkers. Engineers praised the slenderness, the city installed dampers, and nothing says the bridge was too narrow.

Answer explanations

SAT Practice Test 3, Reading and Writing, Module 1

Questions 18 to 33

18 Choice B Inferences

Higher levels near the airport suggest noise may contribute, while the caution means it is not the only factor. The other choices overstate the findings.

19 Choice D Inferences

Speakers must mark the source of every statement and are judged by it, so accuracy about the source matters socially. The text does not support the other options.

20 Choice A Form, structure, and sense

The subject is the plural roses, so the present tense verb bloom agrees. Blooms and has bloomed are singular, and blooming is not a finite verb.

21 Choice A Boundaries

The relative pronoun is the subject of the verb had played, so who is correct, and the clause needs the verb had. The other choices use the wrong pronoun or omit the verb.

22 Choice C Form, structure, and sense

The second clause contrasts with the first, so although is the right conjunction.

23 Choice B Form, structure, and sense

The items in a series must have the same form, so to dive, to swim, and to explore is parallel. The others mix forms.

24 Choice D Boundaries

The possessive pronoun its shows that the volunteers belong to the sanctuary. It's means it is, and their's is not a word.

25 Choice D Boundaries

The subject is the singular report; along with its appendices does not make it plural. The singular verb is agrees.

26 Choice A Form, structure, and sense

The infinitive to win modifies first student. The other choices are ungrammatical.

27 Choice B Form, structure, and sense

The participial phrase eager to see completes the sentence properly. They were eager would create a comma splice, and the others are ungrammatical.

28 Choice C Transitions

The longer running time contrasts with what a smaller battery would suggest, so Even so is the correct transition.

29 Choice D Transitions

The second sentence introduces a different use of honey, so However is a suitable transition in this context. Thus and For example do not fit.

30 Choice C Transitions

The minority view contrasts with the majority, so Still is the best transition.

31 Choice B Rhetorical synthesis

To introduce her work, the choice must say who she is and what she studies, which the first choice does. The others give a workplace, a date or a general fact.

32 Choice A Rhetorical synthesis

The goal is to explain the tool's purpose, so the choice should describe what it measures and how sailors use it, which the first choice does.

33 Choice D Rhetorical synthesis

A comparison needs the shared feature and a contrast between them, which the first choice supplies. The others describe one city or give separate facts.

Answer explanations

SAT Practice Test 3, Reading and Writing, Module 2

Questions 1 to 17

1 **Choice D** Words in context

The court once required a warrant and now does not, so the ruling reverses its earlier stance. Reaffirmation and clarification would keep the position, and postponement does not fit the change.

2 **Choice B** Words in context

Short plain sentences are unadorned, and deceptively signals that the plainness hides effort. Labyrinthine, ostentatious and turgid all describe ornate or complicated writing.

3 **Choice C** Words in context

The contrast with the faded manuscript shows that the vault-stored colours stayed bright, which vivid means. Muted would describe faded colours, and nothing suggests the colours were unstable or ordinary.

4 **Choice A** Words in context

She acknowledged offence without admitting regret, which is evasive. Unreserved and heartfelt describe sincere apologies, and nothing in the passage concerns timing.

5 **Choice C** Words in context

Houses at the same angle and parallel streets show a rigid order, which regimented captures. Haphazard and fortuitous suggest the opposite, and ephemeral means short-lived.

6 **Choice D** Text structure and purpose

The underlined sentence says the doctrine of perfect spheres made the mountainous moon unwelcome, which explains the resistance described. It does not describe the telescope's construction or argue for the doctrine.

7 **Choice A** Text structure and purpose

The sentence turns from Joaquín's childhood memory to the evidence of neglect, from which he infers his father's decline. The text does not blame the agent, give his reason for selling or say when he last visited.

8 **Choice B** Text structure and purpose

The first half describes the speaker's youthful belief that maps were fact, and the second half describes learning to read the signs of doubt. There is no journey, praise of a mapmaker or rule with exceptions.

9 **Choice D** Cross-text connections

Text 2 found employment moved in step across the border, which does not match the fall in jobs that Text 1 predicts. It does not support abolition, and does not discuss pay above the minimum.

10 **Choice B** Central ideas and details

The text reports that rivers in parks without wolves also recovered and urges caution in crediting wolves, so the analysis challenges the assumption. It does not say wolves have no effect on deer, that recovery needs wet years, or that wolves harmed rivers.

11 **Choice A** Central ideas and details

The text says Marchetti rejects the minimalist label because her spareness serves the light, which she treats as her main material. She uses light, the galleries change, and she is described as a minimalist.

12 **Choice C** Command of evidence: quantitative

The claim involves a comparison between the farm lake and the forest lake in summer, which the first choice gives. January figures are not summer, and the other choices ignore the comparison.

13 **Choice C** Command of evidence: quantitative

The claim is about the largest increase, so the correct choice compares the rises. The others report a single year, one exhibition's rise, or a decrease.

14 **Choice D** Command of evidence: textual

Magnifying swings means a self-reinforcing process in both directions, which the last sentence states directly. The other quotations state her thesis or one step in the cycle without showing the full swing.

15 **Choice A** Command of evidence: textual

If fish thrive once placed in a high lake, then conditions there are suitable and only the barrier of waterfalls explains their absence. The other findings do not isolate waterfalls as the cause.

16 **Choice B** Command of evidence: textual

The argument is that summarising helps understanding, and the study found handwriting led to better results on conceptual questions. The groups did equally well on facts, and the text does not compare speed or word counts.

Answer explanations

SAT Practice Test 3, Reading and Writing, Module 2

Questions 18 to 33

18 **Choice C** Inferences

The alloy outlasted steel because it flexes without cracking, so durability depends on tolerance to repeated flexing rather than raw strength. Weight, steel and manufacturing speed are not the basis of the finding.

19 **Choice A** Inferences

A trait that evolves independently many times usually offers an advantage in the environment where it appears. The text rules out a single origin and does not say it occurs in all fish.

20 **Choice D** Form, structure, and sense

The nonessential clause is set off with commas, so the relative pronoun which is needed. That cannot follow a comma in this construction.

21 **Choice A** Boundaries

The subject theory is singular, and the sentence needs a finite verb in the present tense, so is is correct.

22 **Choice D** Form, structure, and sense

After Not only the auxiliary comes before the subject, and the plural subject regulations needs do.

23 **Choice B** Form, structure, and sense

The pronoun stands for the singular interpretation, so that is correct. Those would be plural.

24 **Choice C** Form, structure, and sense

The opening clause explains why the company acted, so the company must be the subject of the main clause. The other choices shift the subject or repeat the reason.

25 **Choice D** Form, structure, and sense

With more than two items, the superlative closest is required. Closer is the comparative, and the others are not idiomatic.

26 **Choice A** Boundaries

The participial phrase having reviewed all of the applications is introductory to the main clause and must be followed by a comma. The other marks would separate it as if it were a complete sentence.

27 **Choice C** Boundaries

The pronoun is the subject of had read, so who is correct.

28 **Choice B** Transitions

The ink dating contrasts with the argument that the manuscript is a forgery, so On the other hand is the logical transition.

29 **Choice C** Transitions

The share price rising contradicts what falling earnings would suggest, so Even so is correct.

30 **Choice A** Transitions

The lower cost of repair is a point in favour of repair that qualifies the bridge's poor condition, so Still is the suitable transition.

31 **Choice D** Rhetorical synthesis

A contrast between findings needs both values, which the first choice supplies. The others describe the journal, the instruments or the dates, not the findings.

32 **Choice B** Rhetorical synthesis

To explain how the process works, the choice must describe the mechanism, which only the first does. The others state a cost, a use or a definition.

33 **Choice A** Rhetorical synthesis

A recommendation supported by research must state what to do and the evidence for it, which the first choice does. The others describe the study, one result or a definition.

Answer explanations

SAT Practice Test 3, Math, Module 1

Questions 1 to 14

1 **Choice C** Percentages

15% of 300 is 0.15 times 300, which is 45.

2 **Choice B** Data analysis

Club 4 has 25 and Club 2 has 23, and 25 minus 23 is 2.

3 **Choice A** Linear equations

Subtract 5x from both sides: $4x - 3 = 41$. Then $4x = 44$, so $x = 11$.

4 **Choice D** Linear equations in context

The fee is paid once, so it is the constant 58. The $23 repeats, so it multiplies m. The total is $23m + 58$.

5 **Choice D** Systems of equations

Subtracting the second equation from the first gives $2y = 14$, so $y = 7$.

6 **Choice B** Functions

Each step of 1 in x adds 6 to f(x), so the slope is 6. Then $f(1) = 7$ gives $6 + 1$, so the intercept is 1.

7 **Choice C** Nonlinear expressions

Multiply the coefficients, 3 times 5 is 15, and add the exponents, 4 plus 4 is 8.

8 **Choice A** Ratios and rates

18 cups of sugar is 9 times 2 cups, so the flour is 9 times 7 cups, which is 63.

9 **Choice C** Triangles

The angles of a triangle add to 180°, so $2x + 3x + 4x = 9x = 180$ and $x = 20$.

10 **Choice D** Circles

The circumference $2\pi r = 10\pi$ gives $r = 5$. The area is $\pi r^2 = 25\pi$.

11 **Choice B** Quadratics

The equation factors as $(x - 4)(x + 2) = 0$, so $x = 4$ or $x = -2$.

12 **Choice A** Probability

49 students travel that way and 21 of them are in Grade 10, so the probability is 21/49, which simplifies to 3/7. The fraction with 117 on the bottom uses every student.

13 **Choice C** Statistics

The first 5 scores total 410, and adding 94 gives 504. Divided by 6, the mean is 84.

14 **Choice B** Scatterplots

Substitute $x = 5$: $3(5) + 54 = 15 + 54 = 69$.

Answer explanations

SAT Practice Test 3, Math, Module 1

Questions 15 to 27

15 **Choice D** Inequalities

The total is the 35 already joined plus 15 each week, and it must be at least 150, so $35 + 15w \geq 150$.

16 **Choice A** Functions

$f(5) = 5(25) - 15 + 5 = 125 - 15 + 5 = 115$.

17 **Choice B** Exponential functions

Doubling every 6 hours means the number of doublings is $t/6$, so the population is 800 times 2 to the power $t/6$.

18 **Choice C** Linear functions

The slope is $(4 - 0)/(4 - 2) = 2$. Using (2, 0), $0 = 2(2) + b$ gives $b = -4$.

19 **Choice D** Percent change

The increase is 1,800, and dividing by the original 12,000 gives 0.15, which is 15%. Dividing by the new value would give a smaller percent.

20 **Choice A** Right triangles

$7^2 + 24^2 = 625$, so the hypotenuse is $\sqrt{625} = 25$.

21 **20** Percent change

The increase is 4 dollars, and 4 divided by the original 20 is 0.2, which is 20 percent.

22 **6** Ratios and rates

The car travels 48 miles per gallon, so 288 miles needs $288 / 48 = 6$ gallons.

23 **396** Volume

The volume is $\pi r^2 h = \pi(36)(11) = 396\pi$, so $k = 396$.

24 **12** Linear equations

Expanding gives $3x - 12 = 6x - 48$. Collecting terms gives $-3x = -36$, so $x = 12$.

25 **23** Systems of equations

Adding $x + y = 34$ and $x - y = 12$ gives $2x = 46$, so the larger number is 23.

26 **63** Area

The area of a triangle is half the base times the height: $(1/2)(18)(7) = 63$.

27 **11** Quadratics

The equation factors as $(x - 11)(x + 6) = 0$, so the solutions are 11 and -6. The positive one is 11.

Answer explanations

SAT Practice Test 3, Math, Module 2

Questions 1 to 14

1 **Choice B** **Volume**

The volume is $\pi r^2 h = \pi(36)(7) = 252\pi$.

2 **Choice A** **Probability**

45 students travel that way and 26 of them are in Grade 10, so the probability is 26/45. The fraction with 106 on the bottom uses every student.

3 **Choice D** **Scatterplots**

Substitute x = 5: 9(5) + 44 = 45 + 44 = 89.

4 **Choice C** **Functions**

f(2) = 4(4) − 12 + 6 = 16 − 12 + 6 = 10.

5 **Choice C** **Statistics**

The first 6 scores total 360, and adding 81 gives 441. Divided by 7, the mean is 63.

6 **Choice D** **Linear equations**

Subtract x from both sides: 4x + 6 = 22. Then 4x = 16, so x = 4.

7 **Choice B** **Percentages**

35% of 240 is 0.35 times 240, which is 84.

8 **Choice A** **Circles**

Arc length is the radius times the central angle in radians: 15 × π/4 = 15/4π.

9 **Choice D** **Exponential functions**

A year is 12 months, so the value is multiplied by 0.7^(12/6) = 0.7^2 = 0.49 each year. That is a decrease of 1 − 0.49, or 51%.

10 **Choice B** **Circles in the plane**

The center of circle A is (4, −4). Shifting it 3 units right and 2 units up moves the center to (7, −2), and the radius is unchanged, so the equation is (x − 7)2 + (y + 2)2 = 36.

11 **Choice A** **Exponential equations**

Write 4 as 2^2, so the right side is 2^(2x − 2). Equal bases give x + 1 = 2x − 2, so x = 3.

12 **Choice C** **Quadratics**

Completing the square: $x^2 - 12x + 27 = (x - 6)^2 - 9$. The smallest value of the square is 0, so the minimum of f is −9.

13 **Choice A** **Systems of equations**

A system has no solution when the lines are parallel but not identical. The second equation is 3 times the left side of the first with a different constant, so a must equal 6/3 = 2.

14 **Choice C** **Absolute value**

The inequality means x is within 3 of 3, so 0 ≤ x ≤ 6. That range has 7 integers, counting both ends.

Answer explanations

SAT Practice Test 3, Math, Module 2

Questions 15 to 27

15 **Choice B** Equivalent expressions

$(4x + 3)^2 = 16x^2 + 24x + 9$, and $(x - 1)^2 = x^2 - 2x + 1$. Subtracting the second from the first gives $15x^2 + 26x + 8$.

16 **Choice D** Ratios and rates

The pumps fill 1/6 and 1/3 of the tank each hour, together 1/2 of it, so the time is 2 hours.

17 **Choice B** Statistics

Multiplying by 2 multiplies the mean and the standard deviation by 2; adding 8 moves the mean by 8 but does not change the spread. The mean of A is 5.6, so the mean of B is 19.2, which is 13.6 greater. The standard deviation of B is 2 times that of A.

18 **Choice C** Equivalent expressions

The numerator factors as $(x - 7)(x - 2)$. Cancelling $x - 7$ leaves $x - 2$.

19 **Choice A** Functions

$g(3) = 10$, so $f(g(3)) = f(10) = 25$. Also $f(3) = 11$, so $g(f(3)) = g(11) = 122$. The difference is $25 - 122 = -97$.

20 **Choice D** Percentages

50 is 30 more than 20, and $30 / 20 = 1.5$, so $p = 150$.

21 **5** Linear equations

Expanding gives $4x - 12 = 8x - 32$. Collecting terms gives $-4x = -20$, so $x = 5$.

22 **-1** Quadratics

For $ax^2 + bx + c = 0$ the solutions add to $-b/a = -(2)/2 = -1$.

23 **10** Quadratics

The maximum is at $t = -b/(2a) = 4/2 = 2$. Then $h(2) = -1(4) + 8 + 6 = 10$.

24 **5** Circles in the plane

Complete the square: $(x + 4)^2 + (y + 3)^2 = 0 + 16 + 9 = 25$. The radius is $\sqrt{25} = 5$.

25 **34** Similar triangles

The scale factor is $DE / AB = 10 / 5 = 2$. The perimeter of ABC is 17, so the perimeter of DEF is $2 \times 17 = 34$.

26 **8** Quadratics

Exactly one real solution means the discriminant is zero: $b^2 - 4(16) = 0$, so $b^2 = 64$ and $b = 8$ (taking the positive value).

27 **-1** Systems with nonlinear equations

Setting the expressions equal gives $x^2 - 4x + (3 - k) = 0$. One solution means the discriminant is 0: $16 - 4(3 - k) = 0$, so $k = -1$.